



2025

X22

BOOKLET NO.

164034

Physics

Paper – I

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

Note : Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.

**SECTION – A**

Q1. Write on the following in about **150** words **each**. **(5×10=50)**

Q1. (a) State and explain Newton's laws of motion. Show that Newton's first law of motion is simply a special case of Newton's second law of motion. Discuss the limitations of Newton's laws of motion. **10**

Q1. (b) State and explain Fermat's principle. Prove the laws of reflection and refraction using Fermat's principle. **10**

Q1. (c) Define Poynting vector and obtain an expression for Poynting vector. **10**

Q1. (d) What are central and non-central forces ? Give three characteristics of each and two examples of each. **10**

Q1. (e) Explain chromatic aberration and obtain relation between the axial chromatic aberration of a lense and the dispersive power of the material of the lense. **10**

Q2. (a) State and prove Kepler's laws of planetary motion. **20**

Q2. (b) State Maxwell's four field equations. Hence, derive electromagnetic wave equation in free space. **20**

Q2. (c) Explain with necessary theory, the formation of Newton's rings in reflected light. Prove that in reflected light, the diameter of dark rings are directly proportional to the square roots of natural numbers and that of bright rings, are directly proportional to the square roots of odd natural numbers. **10**

Q3. (a) Obtain an expression for the gravitational potential due to a thin, uniform spherical shell at a point (i) inside, (ii) at the surface and (iii) outside the shell. **20**

Q3. (b) Obtain the conditions for achromatism of two thin lenses mounted co-axially (i) in contact and (ii) at a small distance. **20**

Q3. (c) An electromagnetic wave is incident normally on a glass surface of refractive index 1.5 with respect to air. Calculate the coefficient of reflection and coefficient of transmission for the wave. **10**



- Q4. (a)** Explain the phenomenon of interference in thin films (i) due to reflected light and (ii) due to transmitted light. **20**
- Q4. (b)** (i) An electromagnetic wave is incident normally on the surface of water of refractive index 1.33. Calculate the percentage of incident energy transmitted. **10**
- (ii) An electromagnetic wave is incident normally on the boundary separating two dielectric media. If the total incident energy is shared equally between the reflected and transmitted waves, find the relation between the refractive indices of two media. **10**
- Q4. (c)** Derive Euler's equation of motion of a rigid body. **10**
-

SECTION – B

- Q5.** Solve the following/Write on the following in about 150 words. **(5×10=50)**
- Q5. (a)** State and explain Kirchoff's current and voltage laws. Using Kirchoff's laws, derive the balance condition of wheatstone bridge. **10**
- Q5. (b)** An A.C. voltage, $E = E_0^{j\omega t}$ is applied to the ends of a circuit containing resistance R, inductance L and capacitance C, connected in series. Calculate (i) the impedance of the circuit, (ii) phase angle and (iii) current at any instant. **10**
- Q5. (c)** Derive Clausius-Clapeyron's latent heat equation. **10**
- Q5. (d)** Explain reversible and irreversible processes in thermodynamics. State the conditions for reversibility and irreversibility. **10**
- Q5. (e)** State and explain : **10**
- (i) Biot – Savart's law
- (ii) Ampere's law
- (iii) Faraday's law
- (iv) Lenz's law.
-
- Q6. (a)** Explain in detail, the construction, theory and uses of a transformer. What are energy losses and how are they reduced to minimum ? **20**
- Q6. (b)** Explain the principle of working of a diesel engine and derive an expression for its efficiency. Discuss the relative merits and demerits of such an engine. **20**
- Q6. (c)** What is self induction ? Derive an expression for coefficient of self induction. State its unit and its dimensional formula. **10**
-



- Q7. (a)** Explain the term resonance as applied to electrical circuits. Find the conditions for sharp resonance and derive expressions for resonance frequency in the case of (i) Series LCR resonant circuit and (ii) Parallel LCR resonant circuit. **20**
- Q7. (b)** Explain the principle of working of Otto engine with the help of an indicator diagram and obtain an expression for its efficiency. **20**
- Q7. (c)** Critical volume of Helium is 3.72 cc/gm. and Van-der-Waal's constant $a = 1.8 \times 10^{-4} \text{ Nm}^4/\text{mol}^2$. Calculate the critical constants of Helium. Molecular weight of Helium = 4 and $R = 8.4 \text{ J/mol-K}$. **10**
-
- Q8. (a) (i)** Two closely wound coils have self-inductances L_1 and L_2 and placed so that practically the entire field set up by either coil, links with all the turns of the other. Compute the mutual inductance M of the coils and show that $M = \sqrt{L_1 L_2}$. Define coefficient of coupling and state its value for no coupling and perfect coupling. **10**
- (ii)** Distinguish between series and parallel resonant circuits. **10**
- Q8. (b) (i)** Using general expression for Maxwell's thermodynamical relations, deduce four Maxwell's fundamental thermodynamical relations. **10**
- (ii)** Show that total entropy of a mole of an ideal gas, having constant molar heat capacities C_p and C_v , is given by – Total entropy, $S = C_v \ln \frac{P_2}{P_1} + C_p \ln \frac{V_2}{V_1}$ where symbols have their usual meanings. **10**
- Q8. (c) (i)** The ratio of number of turns in primary to secondary is 1 : 20. It is connected to a supply of 200 volts A. C. Find voltage across secondary and find the ratio of I_p/I_s . **5**
- (ii)** Calculate the coefficient of mutual induction of a pair of two coils, if a current of 10 Amp. in one coil produces a flux of 10^{-4} Wb . per turn in the second coil of 1000 turns. **5**
-